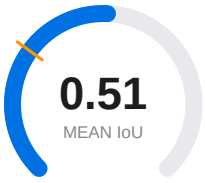
What this document is not. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

FINDING

419792319 is the best physical fit.

Forward replay tested 6 candidates; the runner-up fits at 0.13. A ranking by physical consistency under tested assumptions — an investigative lead for a human analyst.



MEAN IOU	VARIANTS WON	LEAD RETAINED	CANDIDATES
0.51	**6** / 6	**100%**	**6**
the fit gate is 0.3	forcing variants where it fits best	of 80 tested changes	replayed by forward replay

EVIDENCE

- Best fit under **6 of 6** forcing variants
- Ahead of the runner-up by a mean loss gap of **0.1125**
- Inside the origin area **9 h** before the image
- The nearest vessel, **419151108**, 0.0152° away, is not supported

Candidates, ranked by physical fit

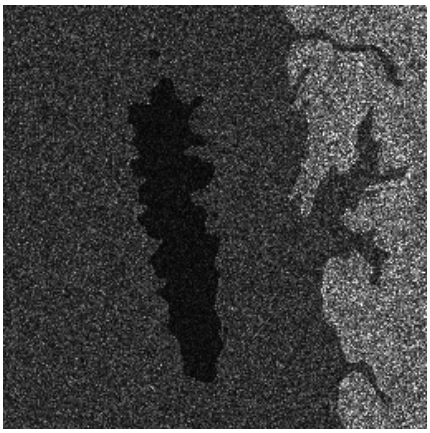
Each candidate's release replayed forward under six forcing variants and compared with the observed slick.

#	VESSEL	MEAN IoU		VARIANTS WON	RELEASE STARTS
1	419792319		0.510	6 of 6	9 Sep 2026, 21:30 UTC
2	419390401		0.130	0 of 6	9 Sep 2026, 04:30 UTC
3	419134716		0.103	0 of 6	9 Sep 2026, 19:00 UTC
4	419578923		0.004	0 of 6	9 Sep 2026, 04:30 UTC
5	419151108		0.050	0 of 6	10 Sep 2026, 04:00 UTC
6	419356707		0.000	0 of 6	9 Sep 2026, 17:30 UTC

The proximity baseline — the vessel nearest the slick when the image was taken — would pick **419151108**, ranked 5 by forward replay.

The scene

The radar image the slick was detected in. Radar darkness alone is not proof of oil.



Acquired	10 Sep 2026, 04:30 UTC
Sensor	synthetic SAR-like (VV proxy, single channel)
Pixel spacing	433.59 m
Data status	synthetic
Label source	generator ground truth (synthetic) — not a human or independently confirmed label
Projection	EPSG:32643

Robustness

Does the leader survive small changes to the model's assumptions?

100%
LEAD RETAINED

80 tested changes to the model's assumptions; **0** changed the leader.

No tested change moved the leader.

Single-axis and two-axis perturbations only. Three-or-more-axis combinations were not tested, so absence of a flip is evidence of stability across the tested set, not a proof of stability.

Forward exposure

Where the oil is modelled to drift after the image. Reanalysis-style retrospective field — not an operational forecast.

AFTER THE IMAGE	MODELLED SPREAD	SHORELINE	FISHING GROUND	MANGROVE RESERVE
+6 h 10 Sep 2026, 10:30 UTC	582 km^2	0%	15%	0%
+12 h 10 Sep 2026, 16:30 UTC	603 km^2	1%	15%	0%
+24 h 11 Sep 2026, 04:30 UTC	616 km^2	4%	14%	1%

Traffic filtering

Every vessel in the scene, checked against where the oil could have come from over the 24 hours before the image.

6 Candidate track data present and inside the backward envelope at some queried time

Unassessed is reported separately from excluded: missing evidence is not exoneration.

Coverage-aware matching

Matching radar contacts to AIS when some AIS is hidden — tested on synthetic contacts.

RATE	VESSELS HIDDEN	AREA GOES DARK
Hidden AIS recovered	**82%** 54/66	**88%** 74/84
False alerts	**27%** 32/118	**19%** 24/127
Mistaken promotions	**15%** 6/40	**14%** 5/36

Provenance

AIS	synthetic (fictional MMSIs; no real broadcast data)
Forcing	slicktrace_core.synth_generator constant+spatial field v0.2
Licence	Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply.
Generated	26 Sep 2026, 17:12 UTC
Software	Python 3.12.14 · Darwin 25.2.0 (arm64)

File integrity

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

attribution_result.json	`351c431307e79375150fdf61da42d3822d10c138c431319d5a0432da2b64a77e`
coverage_result.json	`82c713864f3774fb9a82d8e1cf7a97b15f211244fe4ba6ee85bf009f29568219`
forcing.npz	`75abcfbcc618a9f421ea505279a4490abcc0e188b1c18a4d6f6f47ae52a64b51`
forecast_result.json	`49b1e5d52098088c6a2d6b444d63626a911dda18151d2f1afd57ae9196fc1e52`
fragility_result.json	`d3d9c6722d1c50b28765874dbd2e958584f28853e3a7ef14126299f8c3ae2705`
historical_mapping.json	`ec286ade49884ab142cbc4c6bbccd64bee0875987f4cf7e90cbd651611e2f0ca`
manifest.json	`03333c640d8394098c2fcae739b644ac5299039f9505f8f830b4bb94eb486799`
mask_groundtruth.png	`e924a87a19bb9da27770eeab94e8e8d016a83f24c4fe5f9e99426577ba1bdcf1`
naive_baseline_result.json	`47bab7ddfd6f2887dc4cabe6c0c94ea38c3f2347e46283e9c75a18b85f68f931`
next_observation_result.json	`af7534725417ee1cc6e74fa732ebafa99c2ac84a137dfd519031994112e150d8`
scene.png	`78c3779d73ff851a3db24ace451b69f74cbd1f153d644ab6adf3628818e7f41d`

Uncertainty

- Rankings express physical consistency under tested assumptions. They are not proof of responsibility.
- Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.
- Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.
- Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

Restated. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.